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AI-Enabled Smart Surveillance Systems over 5G Networks

Abstract

Smart surveillance systems are becoming increasingly important for security, public safety, traffic monitoring, and industrial applications. Traditional surveillance systems mainly depend on continuous video recording and centralized processing, which can cause high latency, large bandwidth requirements, and delayed detection of abnormal activities. Artificial Intelligence (AI) can overcome these limitations by automatically analyzing video streams and identifying objects, events, and suspicious activities. When combined with 5G communication and edge computing, AI-enabled surveillance can provide faster, more reliable, and intelligent real-time monitoring.

Introduction

The proposed system combines AI, 5G communication, IoT cameras, and edge computing to develop an intelligent surveillance system. Cameras and sensors continuously collect real-time video and environmental information. The collected data is transmitted through a high-speed 5G network to nearby edge servers. AI models deployed at the edge analyze the video and detect events such as unauthorized access, abnormal movement, accidents, or suspicious activities. Only important information or detected events can be sent to the cloud or control center, reducing unnecessary network traffic.

Proposed Architecture

The system consists of five major components: smart cameras and sensors, 5G communication network, edge computing server, AI-based detection module, and cloud/control center. Smart cameras capture video data and send it through the 5G network. The edge server performs AI-based image and video processing close to the source. Object detection, face recognition, activity recognition, and anomaly detection can be performed using machine-learning or deep-learning models. The results are then sent to the monitoring center for visualization and further action.

Architecture:

Smart Cameras/IoT Sensors → 5G Network → Edge AI Server → Intelligent Detection → Alert/Control Center → Cloud Storage

Role of 5G

5G improves smart surveillance by providing high data rates, low latency, high reliability, and massive device connectivity. High bandwidth allows high-resolution video to be transmitted efficiently, while low latency enables rapid detection and response. 5G also allows a large number of cameras and IoT sensors to operate simultaneously, which is useful for smart cities, airports, hospitals, industries, and transportation systems.

AI and Edge Computing

AI enables the surveillance system to automatically understand video information instead of depending completely on human operators. Edge computing reduces the distance between data sources and processing units, thereby reducing response time. It can also improve privacy because sensitive video data can be processed locally without continuously sending complete video streams to the cloud.

Applications and Benefits

The system can be applied to smart city surveillance, intelligent traffic monitoring, industrial security, public-place monitoring, campus security, airport surveillance, and emergency detection. Major benefits include real-time monitoring, faster threat detection, reduced bandwidth consumption, improved scalability, and reduced dependence on manual surveillance.

Conclusion

AI-enabled smart surveillance over 5G networks provides an efficient approach for real-time security and monitoring. The combination of 5G's low-latency communication, edge computing's fast local processing, and AI's intelligent video analysis can create a reliable and scalable surveillance system. This technology can play an important role in future smart cities and intelligent industrial environments.

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AI-enabled smart surveillance system for secure monitoring and authentication

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The proposed Smart Surveillance System presents a novel, hardware-integrated prototype demonstration aimed The proposed Smart Surveillance System presents a groundbreaking hardware integrated prototype that decisively validates the effectiveness of a dual-branch anti-spoofing model on the low-power edge device, Raspberry Pi 3B+. This prototype goes beyond algorithmic performance research, showcasing a fully functional proof of concept. In contrast to existing surveillance solutions that typically rely on centralized cloud processing or basic recognition systems, our system employs an advanced dual-branch model that utilizes both spatial and frequency-domain features. This approach enables real-time anti-spoofing with an impressive error rate of less than 2%. What truly sets our system apart is its seamless end-to-end integration of cloud-based authentication, edge-level inference for rapid response, and an interactive live video conferencing feature. This configuration empowers immediate verification and action during potential spoofing events. With IoT-enabled devices, our system ensures effortless communication for live streaming, automated alerts, and scalable cloud data management. Coupled with edge computing, it guarantees real-time decision making with minimal latency. Experimental results confirm its high accuracy in distinguishing

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genuine users from spoofing attempts, positioning our solution as a lightweight, proactive, and user-interactive surveillance option that is perfectly suited for homes, enterprises, and public infrastructures.

Keywords Facial recognition, Anti-spoofing, Edge computing, IoT surveillance, Cloud security, Real-time monitoring

Intelligent surveillance systems are a revolutionary invention in security technology. The current surveillance systems are able to provide smarter and flexible security solutions due to the availability of latest technologies like artificial intelligence, machine learning, and deployment to the edge devices due to the supporting technologies like Internet of Things and cloud computing. Instead of relying solely on human supervision and basic motion detection systems, smart surveillance systems are capable of using the real-time data processing and interpretation. These systems include capabilities like facial recognition, behaviour analysis, and anomaly detection by the use of very current and sophisticated technologies and algorithms. By use of pattern and behavior analysis, these technologies allow the system to identify people and forecast potential security issues. As a result, they provide more accurate and pertinent insights. Thus, the use of such a proactive security strategy significantly reduces the probability of breaches, hence improving general safety. Even though smart surveillance is getting upgraded quickly, most current systems still have problems that won't go away soon enough, like high latency because they rely on the cloud, bandwidth bottlenecks in big IoT networks, and being open to spoofing attacks. These are because they don't have enough intelligence locally. Therefore, the suggested hybrid AI–IoT– Edge framework gets around these problems by intelligently dividing up the work between the edge and cloud layers. Edge devices do local facial authentication and anti-spoofing analysis so that decisions can be made right away. The cloud layer takes care of model updates, long-term storage, and system scalability. This architecture reduces latency and protects data privacy by storing the sensitive biometric data locally on the device itself, allowing powerful real-time surveillance in various circumstantial settings. Nonetheless, contemporary hybrid edge–cloud based surveillance systems demonstrate considerable deficiencies, including reliance on singlebranch PAD pipelines, cloud-dependent inference that intensifies latency during high network demand, and insufficient integration of real-time user verification systems. Furthermore, current approaches rarely combine spatial and frequency cues on limited edge devices, which makes them less effective against modern presentation attacks. The highlighted deficiencies need the creation of a lightweight dual-branch architecture incorporating local decision-making and cloud-assisted interaction, as delineated in this paper.

To address these challenges, this paper proposes a hybrid AI–IoT–Edge architecture that performs local processing on edge devices for low-latency decision-making, while leveraging cloud infrastructure for largescale model updates and data management. The objective is to achieve robust and privacy-conscious facial authentication in dynamic environments while minimizing computational load and dependence on centralized servers. Smart surveillance systems are made more capable by the inclusion of IoT devices[1–3]. Cameras, sensors, and other IoT connected devices let these devices interact with one another and with central control systems. Eventually, this results in the creation of a unified network that covers a lot of area and operates perfectly. Using smartphones, tablets, or personal computers, this connection allows consumers to remotely monitor and control their security systems from anywhere in the world. Jameel et al. and Awad et. al. surveyed such Internet of Things (IoT) networks, which enable intelligent and automated systems that are changing many aspects of life. These days the increased utility of IoT devices raises privacy concerns due to the increased information security breaches. The interconnectivity of these devices raises security and privacy concerns, highlighting the need for stronger authentication, encryption, and ethical data handling mechanisms. This paper thoroughly evaluates IoT device authentication systems, including their pros, cons, and security risks.[4,5]

Related works

The smart surveillance systems include a high degree of flexibility and expandability, rendering them appropriate for a diverse array of purposes, spanning from private residences to extensive corporate complexes and public areas. Their services include functionalities such as automated notifications, live video broadcasting, and cloud-based storage for the secure handling of data. These solutions may be tailored to fulfill precise security requirements, guaranteeing that each setting obtains the most suitable amount of safeguarding. Zhang et al. (2023) and others noted that AI-powered facial recognition systems can distinguish real from fake faces[6–8], while Nguyen et al. (2022) and others noted that advanced deep learning algorithms can monitor behavior in real time to identify security risks[9–11]. According to George et al. (2023), IoT integration has enabled real-time monitoring and alerts, reducing reaction times and improving adaptability in varied contexts[12]. Somanathan, S. (2024) stated that IoT-enabled systems may be used in everything from small homes to large industrial complexes due to their scalability[13].

Li et al. (2023) examined how cloud computing improves data management and analytics. This technology also allows predictive security. Cloud computing also raises privacy and moral concerns[14]. Emehin et al. (2024) suggested balancing security and individual rights with robust data protection and transparent processes[15]. Patel S. H.(2024) suggests merging edge computing with 5G technology to analyze and link real-time data and improve security solutions[16]. Prakash et al. (2023) developed a security system that uses Bluetooth, RFID tags, and digital code locks to automatically open doors after authentication[17–19]. The surveillance photos captured in backlit conditions present a significant and complex research challenge in the fields of image processing and computer vision. Yadav et al.[20] use region of interest segmentation and a transfer learning approach to enhance the contrast of dark intensity areas in backlit images using multi-illumination mappings, driven by its practical applications[21,22].

IoT and Raspberry Pi were used to create a home security system by Rani et. al. (2023). The system used WAY2SMS to deliver SMS messages and email illicit photos[23]. Dinakar et al. (2018) developed a Raspberry Pibased IoT home security system. This technology alerts homes about visitors and activates alarms[24]. Anwar et. al. (2016) proposed a home security IoT system with remote door control, smartphone audio alerts, and email notifications for visitor images[25]. Tanaya and Krithiga (2018) installed powerful facial detection equipment to secure their home[26]. Huang et. al. (2022) developed an IoT intercom and chatbot server. Using WoT internet connections, this system controls door locks, cameras, and buzzers. This system enhances safety and convenience.[27] Using sensors and GSM technology, Chandan et al. (2018) created a mobile Internet of Things (IoT) system that can be remotely controlled by smartphones[28]. Face recognition technology was used to create a remote monitoring and response system for automated door locks by Dhobale et al. (2020)[29]. Sheeba et al. developed a motion detection sensor that triggers the doorbell quickly when human movement is detected. A Live Cloud-integrated 24/7 security system communicates with the owner via a microphone when the doorbell is pressed. It alerts the homeowner by comparing the dataset photos to the guest and adding the new visitor image with your authorization using facial recognition technology[30]. Smart surveillance systems are revolutionizing security by using the latest breakthroughs in AI, ML, and the IoT[31–35]. They provide highly precise, streamlined, and reliable security solutions that can easily adjust to the distinct demands of any environment, providing reassurance and improved safety for users around the world.

The proposed system is different from earlier dual-branch PAD models that focus on server-side processing or offline evaluation. It has a frequency-supervised branch and a spatial

MiniFASNet backbone that is made for low-power edge deployment. Additionally, our method is different from previous ones since it includes explicit Fourier Transform loss and real-time cloud-assisted interactive verification. The conceptual differences have been shown in a tabular form in Table 1 above. These features make it possible for IoT, edge, and cloud layers to work together in a more complete way. The schematic layout of the proposed smart surveillance system is represented in Fig. 1. The main contributions of the proposed surveillance system are:

- Surveillance: Integrated AI, ML, IoT, and edge computing for low-latency, real-time monitoring and alerts.
- Authentication: Achieved high-accuracy facial recognition with anti-spoofing, maintaining error rates below 2%.
- Interactive System: Enabled cloud-based scalability and interactive features like live video conferencing using AWS and Jitsi.

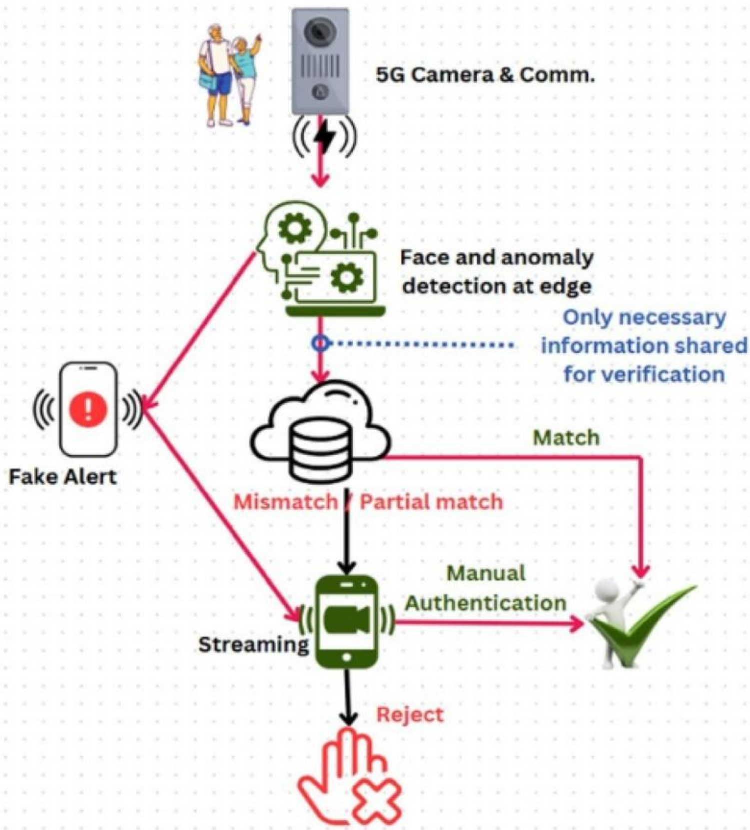


Fig. 1. Schematic layout of the Smart Surveillance System showing AI-driven analytics integrated with IoT cameras, edge nodes, and cloud authentication.

Applications: Designed for diverse use cases–smart homes, offices, and public spaces–enhancing security across environments.

The rest of the paper is organized as follows: Section 2 presents operational flowchart and proposed system model. Section 3 discusses the proposed methodology. Section 4 discusses the integration of hardware-software setup and core functionality of the system, along with experimental results and analysis for performance evaluation. Finally, the authors have concluded the discussion of the system in Section 5.

Proposed system

The operational flow outlines a facial recognition access control system, clearly showing the steps from receiving a live video feed to allowing or denying access.Initially, the system acquires a real-time video stream and preprocesses each frame for face detection, including noise reduction, feature enhancement, and normalization techniques. Upon successful face detection, the system extracts critical features and conducts a comparison against a pre-stored database of authorized individuals. If a match is identified, the system verifies the individual's identity and grants access in accordance with their predefined privilege level.

If the detected face is not in the database, it is an unidentified visitor. In such cases, the system activates a security alert to inform the premises owner of the visitor's presence. Additionally, live video calls can be initiated to facilitate real-time verification. This operational flow, as shown in Fig. 2, establishes a seamless yet secure access control mechanism, integrating real-time monitoring, automated identification, and security alert functionalities. The system employs advanced computer vision, artificial intelligence, and robust security protocols to enhance both accuracy and responsiveness in access management.

The schematic layout of the proposed system model explaining the working principle of the "Smart Surveillance System" has been shown in Fig. 1. The advanced surveillance system integrates 5G connectivity, artificial intelligence, cloud computing, and edge processing for effective real-time monitoring in dynamic security environments. It begins with a 5G-enabled security camera, configured for user registration and device integration, initiating a complex authentication process linked to a management dashboard. Facial authentication data is securely sent to the cloud, where deep learning enhances AI models for improved facial recognition and anomaly detection. As the system evolves, trained AI models are deployed to edge devices, allowing for immediate authentication and reduced latency. Interactions with the cloud layer are limited to authenticated users, and sensitive biometric information is stored locally on the edge device to safeguard user privacy. Parameters of the cloud model and system scripts are safeguarded within access-controlled repositories,

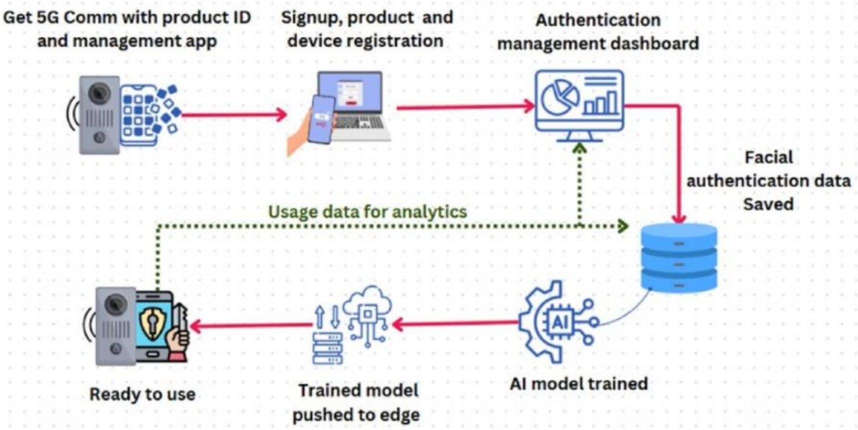


Fig. 2. Operational flowchart of the Smart Surveillance System outlining face detection, anti-spoofing, cloud verification, and real-time alert response.

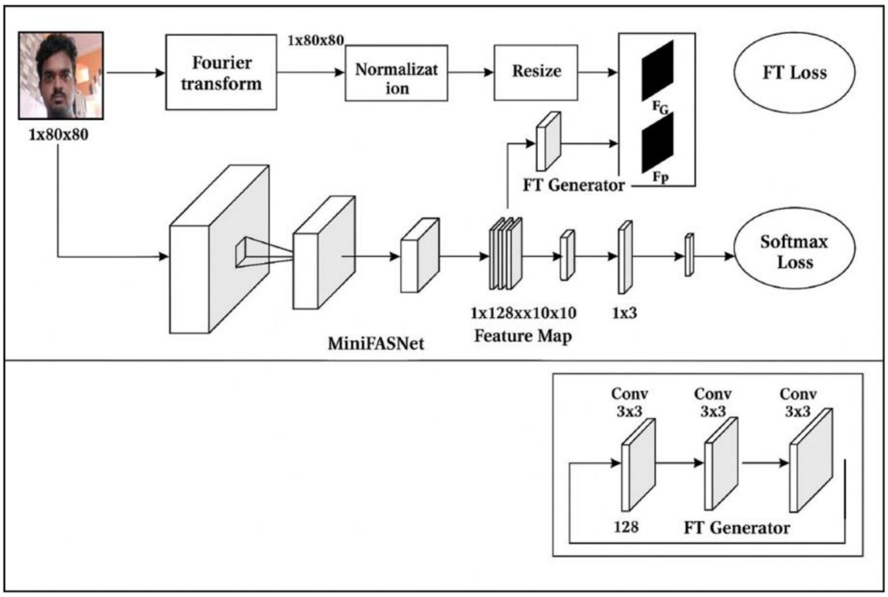


Fig. 3. Dual-branch architecture for face anti-spoofing combining spatial and frequency-domain feature extraction for robust attack detection.and the connection between edge devices and cloud services is secure. This work does not employ cryptographic protocols; rather, these methods regulate access and provide secure system functionality.

Methodology

To address the challenge of presentation attack detection (PAD) in face recognition systems, the proposed architecture employs a dual-branch structure as shown in Fig. 3 designed to leverage both spatial and frequency domain information for robust face anti-spoofing. This design facilitates comprehensive feature extraction by jointly learning from facial appearance and spoof-specific frequency artifacts, enhancing the system's ability to detect various types of presentation attacks[36]. This proposed architecture incorporates anti-spoofing with MiniFASNet, a lightweight convolutional neural network, and Fourier domain supervision. The integration of these two modalities enhances the robustness and generalizability of the anti-spoofing model, especially in realworld scenarios where attack variations can be substantial.

Architectural design rationale

Mini FASNet was chosen as the spatial backbone because it is light and has been shown to work well on mobile and embedded devices with limited computational capacity. The depth-wise separable convolutions and compact channel structure, which can include anywhere from 16 to 128 channels per block, make it possible to do real-time inference on hardware that is similar to a Raspberry Pi without needing GPU acceleration. This architectural choice makes it possible to maintain a constant frame rate while yet being able to handle PAD demands. The dual-branch architecture strikes a balance between processing performance and discriminative capabilities by making sure that both branches create feature maps with the same spatial dimensions (10×10). This makes fusion possible.

Frequency domain branch

The upper branch of the proposed architecture is dedicated to capturing spoof-specific patterns in the frequency domain. The input to this branch is a grayscale facial image of dimension 1 × 80 × 80. Initially, a Fast Fourier Transform (FFT) is applied to project the image from the spatial to the frequency domain, revealing periodic structures and noise patterns that are often introduced by spoofing media such as printed photographs, replayed videos, or digital screens.

Given an input grayscale image $I \in R^{1 \times H \times W}$ W where H=W=80, and first of all a 2D Discrete Fourier Transform (DFT) has been performed to obtain the frequency spectrum:

$$F(u, v) = \sum^{H-1} \sum^{W-1} I(x, y) \cdot e^{-2\pi i\left(\frac{ux}{H} + \frac{vy}{W}\right)} \tag{1}$$

$x=0\ y=0$ where $F \in C^{H \times W}$ is the complex-valued frequency map. The magnitude spectrum is then computed as:

$$F(u,v)| = \sqrt{\mathrm{Re}(F)^2 + \mathrm{Im}(F)^2} \tag{2}$$

This magnitude map is normalized and resized to a fixed dimension $F_{\text{resized}} \in R^{1\times10\times10}$ using bilinear interpolation to form a compact frequency representation. The 10×10 dimensionality was utilised to make sure that the spectrum representation was small enough to fit on the Raspberry Pi 3B+ and that it matched the output resolution of the spatial branch while being less likely to pick up high-frequency noise. The processed frequency tensor is passed through the FT Generator, a lightweight CNN module convolutional layers with kernel size 3×3 and decreasing channel dimensions $\varphi_{FT}((128 \cdot)$, composed of three successive $\rightarrow 64 \rightarrow$ output).

$$F_p = \emptyset_{FT}(F_{\text{resized}}) \tag{3}$$

The FT Generator extracts a predicted frequency representation F_P , which is trained to approximate a ground truth frequency template F_G. A dedicated Fourier Transform (FT) loss function is employed to minimize the discrepancy betweenF_P and F_G, encouraging the network to learn discriminative frequency-based features associated with spoofed content. To ensure that the FT Generator learns meaningful frequency cues, we define a frequency transformation loss L_{FT} as the Mean Squared Error (MSE) between the predicted map FP and a predefined ground truth frequency map F_G:

$$L_{FT} = \frac{1}{N}\sum_{i=1}^{N} \| F_{p(i)} - F_{G(i)} \|_2^2 \tag{4}$$

Where, N denotes the batch size.

Spatial domain branch

The lower branch focuses on learning spatial features through direct processing of the original grayscale image using a compact yet effective backbone, MiniFASNet to extract high-level semantic and textural features directly from the spatial representation of the input image. This network architecture is designed for low-latency deployment and consists of multiple convolutional blocks tailored to extract hierarchical spatial features. The input image is progressively encoded into a deep feature map of size $1_{C\ H\ W} \times 128\times10\times10\varphi$, encapsulating textural, $_{\text{spatial}}(\cdot)$ encodes the structural, and contextual information critical for spoof detection. The network image into a compact feature map connected layer to generate a 3-class prediction vector.These features are further compressed through additional $M \in R^{\times\times}$, which is then globally pooled and passed through a fully convolutional and fully connected layers to form a 1×512 feature vector. A subsequent classification layer maps replay attack. The spatial branch is trained using a conventional softmax cross-entropy loss, which ensures this vector to a 1×3 output, representing the posterior probabilities of three classes: genuine, print attack, and accurate multi-class classification based on spatial features alone.

$$\hat{y} = \text{Softmax}\left(W.\text{Flatten}\left(\varnothing_{\text{spatial}}(I)\right) + b\right) \tag{5}$$

Where $\hat{y} \in \mathrm{R}_3$ denotes the predicted probability distribution over the classes: genuine, print, replay. W and b represent the learnable weight matrix and bias vector of the fully connected classification layer, respectively, while y denotes the one-hot encoded ground truth label corresponding to the three output classes (genuine, print attack, and replay attack). The spatial branch is supervised using the standard cross-entropy loss:

$$L_{CE} = -\sum_{k=1}^{3} y_k \log \hat{y}_k \tag{6}$$

where $y \in \{0,1\}^3$ is the one-hot encoded ground truth label. **Joint optimization and synergistic learning**

The two branches are trained jointly within a multi-task learning framework, where both the FT loss and the softmax loss are simultaneously optimized. This collaborative training paradigm allows the network to learn rich and complementary feature representations. The frequency domain branch improves the model's sensitivity to spoofing artifacts often invisible in raw spatial data, while the spatial domain branch reinforces the learning of contextual and structural features characteristic of genuine faces.

$$L_{\text{Total}} = \lambda_1 L_{CE} + \lambda_2 L_{FT} \tag{7}$$

where λ_1 and λ_2 are scalar weights that balance the contribution of each loss component. In our implementation, these weights are empirically set to emphasize both classification accuracy and spectral feature learning.

The classification is restricted to genuine, printed, and replay attacks, as these represent the primary threat vectors in residential and commercial access control environments. For reliable detection, mask-based 3D attacks usually need depth sensors or infrared-assisted photography. The cheap 2D USB camera used in this prototype can't do either of these things. Integrating mask assaults into the model is acknowledged as a potential direction for future research. This proposed architecture has two parallel branches, with the upper branch representing frequency domain conversion. The Fourier transform is employed for frequency domain conversion, transforming real-time images from the spatial domain to the frequency domain. Subsequently, normalization techniques are employed to scale the variables, ensuring stable training for the model. The images are resized by down-sampling. The FT generator facilitates the generation of a comparison between fake and authentic features, enabling the model to acquire valuable frequency features. To enable the model to make predictions. Likewise, the foundational branch architecture, miniFASNET, a lightweight CNN model, extracts spatial characteristics such as texture and reflections. This architecture use convolutional layers to compress the map into a flat feature vector of 1x512. This feature vector can be utilized in the classification layer, where the model employs softmax loss to oversee the classification process.This dual-branch architecture not only improves the PAD performance across diverse attack types but also maintains computational efficiency, making it well-suited for real-time applications on resource-constrained platforms such as mobile devices, embedded systems, or edge AI environments. By leveraging the synergy between spatial and frequency information, the proposed framework presents a comprehensive and scalable solution for robust face anti-spoofing in practical deployment settings.

Ethics approval and informed consent

All methods were carried out in accordance with relevant guidelines and regulations. This study was conducted as part of the academic curriculum for B.Tech Computer Science and Engineering students during their finalyear project. The experimental protocols were reviewed and approved by the Departmental Project Evaluation Committee of the Department of Computer Science and Engineering, ITER, Siksha 'O' Anusandhan (Deemed to be) University, along with the supervising faculty member.

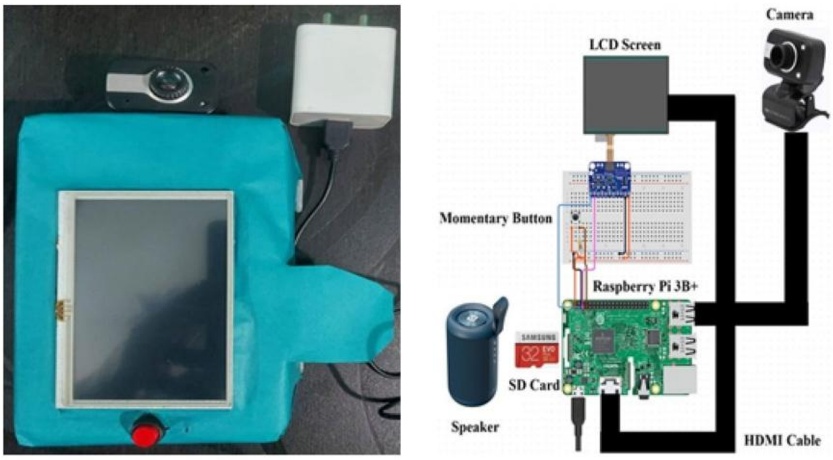
Written informed consent was obtained from all participants prior to data collection and experimentation. Participants were informed about the purpose of the study, the use of their facial images for research and publication, and their right to withdraw at any stage without any consequences.

Experiment and discussion

This section demonstrates the hardware-software setup and functionality using a Raspberry Pi 3B+-based setup that runs a Dual-Branch Model for real-time face recognition and spoof detection using spatial and frequency features. It uses a USB camera, LCD, speaker, and SD card for input, output, and storage–enabling low-power, edge-level biometric verification. The method was evaluated using a restricted private dataset acquired directly from the Raspberry Pi 3B+ camera under standard indoor settings. The collection includes real user attempts, print-based attacks, and replay attacks shown on mobile screens. We did not include public datasets like CASIAFASD and OULU-NPU since their imaging conditions are very different from those of embedded edge devices. For example, they use high-quality sensors, controlled lighting, and static backgrounds, while embedded edge devices use low-resolution, changing lighting. Since the goal of this study is system-level validation on a real deployment platform, using datacollected by Pi is a better way to show how the application is supposed to work. A limited number of consenting individuals supplied facial samples for the private dataset of this investigation under authentic indoor deployment conditions. Between six and ten participants provided several samples from three categories: genuine access attempts, print-based spoofing attacks, and mobile screen replay assaults. A USB camera linked to the Raspberry Pi 3B+ acquired photographs in domestic and professional environments with diverse lighting conditions and backdrops. The dataset confirmed system-level operation, rather than statistical generalizability.

Integrated hardware-software set-up

The hardware-software setup shown in Fig. 4a & b is specifically designed to operationalize the Dual-Branch Model for real-time face recognition and spoof detection on an embedded platform. At its core, the Raspberry Pi 3B+ serves as the computation unit, executing the two parallel branches of the model, one for extracting spatial features using MiniFASNet and the other for processing frequency features via Fourier transform. The connected USB camera captures live facial inputs, which are simultaneously fed into both branches of the model running locally on the Pi. The momentary push button initiates the recognition cycle, triggering the model inference process. The Raspberry Pi 3B+ had an inference time of roughly 250–300 ms per frame during deployment. This



(a) (b)

Fig.. a The stand-alone hardware **b** Internal hardwired connectivity of the hardware

meant that the processing latency was about 0.4 seconds, which included image acquisition and preprocessing. This performance was good enough for real-time access control applications. Once the spatial and frequency features are extracted and evaluated through the dual-loss mechanism (Softmax Loss and FT Loss), the output is displayed on the LCD screen, while the speaker provides real-time audio feedback. The SD card stores the model weights and scripts, enabling low-power offline operation. The hardware components of this integrated hardware-software setup that enables seamless deployment of the dual-branch architecture, demonstrating its practicality for edge-level biometric verification with antispoofing capability, have been explained as follows:

Raspberry Pi 3B+ (central processing unit)

The Raspberry Pi 3B+ acts as the main computational engine that runs the facial recognition algorithm and the Fourier-based spoof detection model. It Interfaces with peripheral components and handles the data processing pipeline in real-time. The pretrained deep learning model, built using frameworks like TensorFlow or PyTorch and optimized with tools like TensorFlow Lite, is deployed on this device. It also manages I/O operations, face detection routines, image acquisition, and prediction logic.

Camera module (image acquisition unit)

An external USB webcam is connected to the Raspberry Pi and is responsible for capturing live facial images. These images are fed into the recognition pipeline where they are simultaneously analyzed by the MiniFASNetbased feature extractor and Fourier Transform generator as discussed previously. This camera initiates the verification cycle whenever triggered by an external input.

LCD display (visual feedback interface)

The LCD screen provides visual feedback of the recognition result. It displays the captured image with bounding boxes and identity information. In the case of a spoof attempt or unknown user, appropriate warnings or messages are shown on the screen. This makes the system suitable for interactive applications like automated check-ins, secure door entry systems, or exam proctoring interfaces.

Momentary push button (trigger mechanism)

The momentary button acts as a manual trigger to initiate the image capture and recognition process. This prevents constant image acquisition, reducing computational load and power consumption. It introduces a level of control to the system, making it user-friendly for authentication tasks.

Speaker (audio feedback unit)

The speaker connected to the Raspberry Pi is used to deliver audio-based feedback. Depending on the outcome (e.g., “Access Granted”, “Access Denied”, “Spoof Detected”), pre-recorded or synthesized voice messages are played. This audio-visual integration enhances the system’s interactivity and accessibility, especially in environments with visually impaired users or hands-free operational needs.

SD card (storage and boot medium)

A 32 GB microSD card is used to store the Raspberry Pi OS, the pre-trained deep learning models, and Python scripts necessary to control the system. It also retains logs of recognition events for further evaluation or audit.

HDMI interface (extended display support)

An HDMI cable enables the output to be extended to an external monitor, useful during the development phase or when more detailed logs and debugging visuals are required.

This hardware supports the deployment of the dual-branch architecture where: the spatial domain branch (MiniFASNet) extracts feature maps from captured RGB images, the frequency domain branch applies the Fourier Transform to detect spoofing patterns and finally, both branches contribute to the final decision via FT loss and Softmax loss, as shown in the earlier model flowchart. The compact and modular design of this hardware ensures that the proposed model is not limited to cloud-based infrastructure, but can also function independently on edge devices, making it ideal for low-resource and offline environments.

Functionality and analysis

The proposed intelligent surveillance system combines deep learning-based facial recognition, anti-spoofing, and real-time video conferencing to ensure secure access control. It operates effectively across three key scenarios: authenticating genuine users, detecting and blocking spoof attacks using photos or screens, and identifying and rejecting unregistered individuals. The system demonstrated high accuracy, distinguishing real faces from spoofed or unknown inputs and ensuring uninterrupted communication while maintaining robust security through low Attack Presentation Classification Error Rate (APCER)[37] and Bona Fide Presentation Classification Error Rate (BPCER)[38,39] values.

$$APCER = \frac{FP}{(TN + FP)} \tag{8}$$

$$BPCER = \frac{B}{N_b} \tag{9}$$

$$FRR = \frac{Numberoffalserejections}{Totalnumberofgenuineattempts} * 100\% \tag{10}$$

$$UDR = \frac{NumberofUnknownAttacks/EventsDetected}{TotalNumberofUnknownAttacks/Events} * 100\% \tag{11}$$

Face detection and recognition process

The suggested facial anti-spoofing and intelligent surveillance system efficiently amalgamates deep learningdriven facial recognition, anti-spoofing methodologies, and real-time video conferencing to augment security. The validation method encompasses many phases of testing and assessment, guaranteeing that the system effectively distinguishes between authentic human faces and spoofing efforts, while facilitating uninterrupted video communication among users. To validate the efficacy of the proposed face recognition and anti-spoofing system, it was tested across three distinct operational scenarios: genuine user identification, spoof attack resistance, and handling of unregistered faces. Each scenario addresses a fundamental capability required for a secure and reliable biometric system. This process begins with the activation of the face detection module by transforming the input video frames from BGR to RGB format via $cv2.cvtColor(frame, cv2.COLOR_{B}GR2RGB)$. This change is crucial as the $face_{r}ecognition$ library functions solely with RGB photos. The $face_{l}ocations$ function utilizes a pre-trained deep learning model to detect and delineate bounding boxes around faces inside the frame.

Genuine user identification

In case of a genuine live face detection condition as shown in Fig. 5, a live face input from a user already enrolled in the database is provided to the system. The model successfully detects and matches the face to the correct identity with high confidence representing by a green bounding box. The True Acceptance Rate (TAR) for this condition is measured as the ratio between the number of correctly identified genuine users to the total number of genuine attempts which was found to be TAR of 98.7% as shown in Table 1, indicating a high level of accuracy in recognizing enrolled users under varying illumination and background conditions. The small number of examples in the prototype assessment dataset makes it impossible to use statistical graphs like ROC curves, EER studies, or full confusion matrices to get useful generalisation insights. We show TAR, APCER, BPCER, FRR,

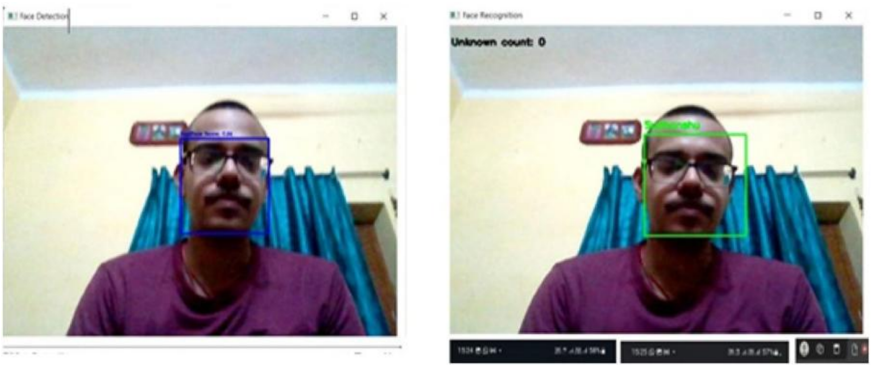


Fig.. Test case for face Detection and Recognition Process by the proposed

Approach	PAD modality	Edge deployable	Frequency supervision	Cloud/IoT integration	Real-time interaction
Traditional CNNbased PAD	Spatial only	Limited	No	No	No
Dual-branch PAD (prior works)	Spatial + frequency	Not optimized	Partial/ implicit	No	No
Proposed System	Spatial + explicit FFT loss	Yes (Raspberry Pi 3B+)	Yes	Yes	Yes (Jitsi video call)

Table 1. Conceptual comparison of proposed system with representative PADenabled surveillance approaches.

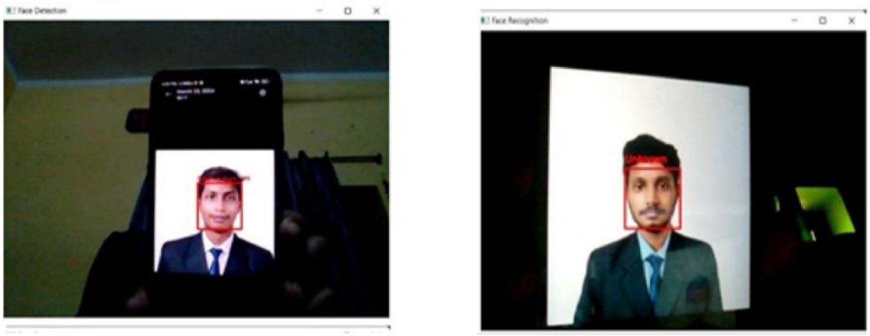


Fig.. Test case for Spoof attack

and UDR, which are better for testing in small real-world situations. This study emphasizes real-time system validation on resource-limited hardware, thereby excluding cross-validation and independent training trials.

Spoof attack resistance

The second scenario is about the detection of a test case of spoof attacks, as shown in Fig. 6, the system is presented with high-quality printed or digital photographs of the enrolled individuals, intended to spoof the system. Despite the visual resemblance to real users, the system correctly identifies these attempts as fraudulent. Figure 5 below with red bounding box shows such cases, where spoofed faces were flagged or not granted access. To evaluate the robustness of the anti-spoofing mechanism, the APCER and BPCER can be determined using Eqs. 8 and 9, respectively. Table 1 depicts the CER of the proposed dual-branch architecture, found to be 1.2% and 1.6% respectively, thus demonstrating high spoof resistance and minimal rejection of genuine users. We didn't directly compare heavy PAD systems like CDCN, DeepPixBiS, or FaceBagNet because they need a lot of processing power and GPU acceleration, which makes them unsuitable for real-time use on Raspberry Pi-class hardware. The focus of the proposed study is on edge-deployable PAD functionality instead of achieving cuttingedge accuracy on large benchmarks. Besides validation demonstrates high accuracy in distinguishing genuine faces from spoofing attempts, with an overall classification error (mean of APCER and BPCER) under 2%. Limited computer resources hindered retraining and recurrent evaluation cycles on Raspberry Pi-class hardware, obstructing a quantitative baseline comparison with a spatial-only MiniFASNet model. The contribution of the frequency-supervised branch is elucidated through architectural design rationale and a defined Fourier-domain loss formulation, enhancing sensitivity to spoof-specific aberrations that spatial clues fail to detect. This method prioritizes edge deployment and robustness.

This validation guarantees that the system is impervious to presentation attacks using printed graphics or digital screen displays. Using convolutional layers, max pooling, and dropout regularization, the model improves its ability to reject fraudulent authentication attempts, rendering the system exceptionally reliable for practical applications.

Handling of unregistered faces

In the last condition, an unknown but real individual, whose face is not present in the enrolment database, is presented to the system. As depicted in Fig. 7 with a limit box labeled 'Unknown' or 'Stranger', the system correctly classifies these inputs as unregistered, thus denying access. This showcases the system's capability for maintaining strict access control. The performance in this scenario is evaluated using the False Rejection Rate (FRR) defined as the ratio between Number of genuine users incorrectly rejected to the total number of genuine attempts as shown in Eq. 10.

Furthermore, Unknown Detection Rate (UDR), which is again defined as the ratio between the total unknown attempts to the Number of unknown faces correctly rejected as shown in Eq. 11. Table 2 shows that the system recorded an FRR of 1.3% and a UDR of 99.1%, highlighting its effectiveness in distinguishing between enrolled and non-enrolled individuals (Table 3).

Video conferencing and surveillance integration

The solution integrates facial recognition with a React-based surveillance platform that employs Jitsi Meet for real-time video communication. The JitsiMeet Component is configurable with parameters like as room names,

Fig.. Test case for "Unknown" or "Stranger" based on unregistered

Test Case	TAR	APCER	BPCER
Overall	**98.7%**	**1.2%**	**1.6%**

Table 2. Observation for true acceptance rate and classification error rates (CER) for user identification and spoof attack detection.

Test Case	FRR	UDR
Overall	**1.3%**	**99.1%**

Table 3. Observation for False Rejection Rate (FRR) and Unknown Detection Rate (UDR).

domains, and user credentials, facilitating the customization of security protocols, audio/video configurations, and interface choices. The system handles participant information via user Info objects, ensuring accurate identification during meetings. In case the system finds the image of an "Unknown" or "Stranger", then a live video call gets enabled by the system. Simultaneously, a security alert is activated, informing the homeowner about the presence of a visitor at it's front door. The Jitsi Meet interface offers a direct video connection, facilitating immediate verification and engagement. The seamless amalgamation of facial recognition with video conferencing guarantees augmented security, remote authentication, and an increased user experience.

Limitations

While the proposed intelligent surveillance system demonstrates promising functionality in controlled conditions, several limitations should be acknowledged. The study is limited by the use of a small Pi-captured dataset rather than larger public PAD datasets, and cross-database analysis was not possible because of a lack of resources. Also, it was not possible to do ROC/EER benchmarks and training comparisons with bigger designs on the hardware platform that was chosen. Mask-based 3D attacks were stopped since they need sensing modalities that the current camera setup doesn't provide. These components represent intrinsic additions for future iterations of the system. The accuracy of facial recognition and anti-spoofing mechanisms may be affected by environmental factors such as variable lighting, background complexity, and camera placement. Additionally, the training dataset may not sufficiently capture the full range of demographic diversity, which could influence the system's performance across different populations. The reliance on cloud-based infrastructure introduces a dependency on stable internet connectivity, potentially limiting usability in low-connectivity environments.

Conclusion

The proposed intelligent surveillance system integrates AI-driven facial recognition, anti-spoofing techniques, and real-time video conferencing into a modular and efficient security framework. Built using Python, PyTorch, OpenCV, React.js, AWS Amplify, and the Jitsi Meet React SDK, the system performs identity verification, spoof detection, and remote interaction functionalities to enhance access control and reduce unauthorized entry attempts in environments such as smart homes, office spaces, and public monitoring systems. While the current implementation demonstrates promising results in controlled environments, further testing in real-world, large-scale deployments is required to assess performance consistency, scalability, and robustness under varying operational conditions. However, the system faces practical deployment challenges, including dependency on stable internet connectivity for cloud-based operations, latency in video streaming during high network load, and reduced accuracy in non-uniform lighting or dynamic crowd settings. Addressing these aspects in future iterations will enhance its adaptability and reliability across diverse use cases.

Data availability

Data Availability: The datasets generated and/or analysed during the current study are available from the corresponding author on reasonable request.

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Author contributions

FAA: Conceptualization, Methodology,Writing- Original draft preparation, and Editing. SM: Validation, Reviewing, and Editing. RM: Conceptualization, Methodology, Writing- Original draft preparation, and Editing. GY: Investigation, Reviewing, and Editing. All authors reviewed the manuscript.

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Consent to publish

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Additional information

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